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MAJOR ARTICLE

Effectiveness of Seasonal Influenza Vaccination Against Medically Attended Influenza in People With Chronic Respiratory Diseases: A Multicenter, Test-Negative, Case-control Study

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Background. Patients with chronic respiratory disease (CRD) face an increased risk of severe influenza complications. However, limited studies offer estimates of vaccine effectiveness (VE) against influenza within these CRD populations in China.

Methods. A multicenter, retrospective, test-negative, case-control study was conducted to estimate the VE in 37 medical institutions in Shanghai, China, during the 2023/2024 and 2024/2025 seasons. We included patients with CRD who presented with acute respiratory infections and received nucleic acid amplification tests and/or rapid antigen tests. Patients with CRD with a positive test were assigned to the case group, and those with a negative test were assigned to the control group. Multivariable unconditional logistic regression was used to control potential confounders and to determine 95% confidence intervals (CIs).

Results. A total of 10 711 participants, including 1650 influenza cases (5.8% vaccinated) and 9061 influenza-negative controls (7.9% vaccinated), were eligible for analysis. The combined VE over the 2 seasons was estimated to be 44.23% (95% CI: 30.45-55.75) for the study population. The VE was 42.91% (95% CI: 27.13–55.78) in the 2023/2024 season and 51.51% (95% CI: 17.70–73.61) in the 2024/2025 season. The combined VE for influenza subtypes A and B were 38.49% (95% CI: 21.75–52.27) and 62.64% (95% CI: 39.91–78.28), respectively.

Conclusions. Influenza vaccination provides consistent and moderate protection to patients with CRD against medically attended influenza, regardless of the dominant circulating subtypes. Nevertheless, vaccination coverage remains suboptimal, underscoring the need to improve annual influenza vaccination uptake among patients with CRD , even amid ongoing viral antigenic evolution through shifts and drifts.

Keywords. chronic respiratory diseases; seasonal influenza; test-negative design; vaccine effectiveness.

Seasonal influenza is an acute respiratory infection (ARI) caused by the influenza viruses [1]. The influenza virus subtypes A/H1N1, A/H3N2, and B/Victoria circulate among humans worldwide and lead to a huge disease burden [2]. Globally, influenza causes 3 to 5 million severe cases and 290 000 to 650 000 respiratory deaths annually. Patients with medical conditions like cardiovascular or chronic respiratory diseases (CRDs) face a higher risk of severe illness or complications [3–5].

In China, the implementation and relaxation of public health and social measures against the coronavirus disease 2019 (COVID-19) over the past 5 years have altered the seasonal circulation patterns of influenza [4–7]. In late 2023, ARIs in China significantly increased compared to pre-COVID seasons, due to high influenza activity and co-circulation of various respiratory pathogens [8]. The pandemic raised public awareness of infectious disease control and prevention [9] and changed people's health behaviors, affecting influenza vaccination rates and willingness [10]. For example, the influenza vaccination rate during the 2021/2022 season (17.68%) in Shanghai was higher than that during the 2018/2019 season (11.8%) [10].

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Influenza vaccine effectiveness (VE) varies across different influenza seasons, populations, levels of vaccine coverage, and the specific outcomes under study [11]. In the 2023/2024 season, the VE against influenza among the general population was 49.4% in Ningbo (Southern China) [12] and was 44.8% in Beijing (Northern China) during the same period [13]. Although previous studies on influenza VE often include participants with underlying chronic diseases, limited studies offer estimates of VE against influenza within these high-risk populations, particularly among individuals with CRDs.

This gap is concerning given that CRDs are among the most common non-communicable diseases globally [14]. However, CRDs have received less public attention when compared to cardiovascular disease, cancer, stroke, and diabetes [15, 16]. Moreover, previous studies showed that individuals with CRDs infected with the influenza virus were more likely to develop pneumonia or experience prolonged and complicated hospitalizations, such as the need for mechanical ventilation, intensive care unit admission, or in-hospital death [17–19]. This causes a considerable burden on the Chinese healthcare system [20]. Vaccination with the influenza vaccine is the most effective way to reduce acute exacerbation, influenza-related ARIs, hospitalizations, and deaths for individuals with CRD [21]. However, previous studies showed that the vaccination rate was lower among people with CRDs [22, 23]. Since influenza vaccination is a public health recommendation for those with CRDs, there is an urgent need to accurately estimate the VE against influenza infection in patients with CRD [24]. Furthermore, most national immunization committees rely on evidence from observational studies rather than randomized controlled trials to evaluate influenza VE [11]. Among observational designs, the test-negative design (TND) is widely regarded as the gold standard for minimizing bias in influenza VE estimation [25, 26]. Therefore, we employed a TND case-control study to estimate VE for each influenza season.

Our study aimed to assess influenza VE against medically attended influenza among patients with CRDs during the 2023/2024 season and the 2024/2025 season.

METHODS

Data Source

The patients' with CRD medical records were based on data from the Pudong New Area Health Big Data Platform to obtain information from medical institutions. The Pudong New Area Health Big Data Platform covers around 60% of outpatient and emergency visits and 41% of inpatient visits in Pudong New Area. The Pudong New Area, Shanghai's largest and most populous district in subtropical southeast China, had a census population of 5.7 million in 2024. Medical records of outpatient, inpatient, and emergency settings in medical institutions come from 7 general hospitals, 11 specialized hospitals, and 48 community health service centers in Pudong. The list of institution codes and names included in this study is shown in the Supplementary Table 1.

The vaccination information data was derived by linking the immunization registry records in Shanghai using the identification number.

Influenza sentinel surveillance data of Pudong were obtained from weekly reports in the Chinese National Influenza Center (CNIC) [27]. Each surveillance week, the CNIC releases the number of influenza-like illness (ILI) specimens and isolated influenza strains from ILI respiratory specimens, determined by nucleic acid amplification tests (NAATs).

Study Design and Study Population

A multicenter, retrospective, test-negative, case-control study was used to estimate the influenza VE for preventing ARIs among medically attended patients with CRD for the 2023/2024 and 2024/2025 influenza seasons. The patients with CRD were identified by using the standardized sets of International Classification of Diseases 10th revision codes (ICD-10) of the Pudong New Area Health Big Data Platform. CRDs include diseases such as COPD, asthma, interstitial lung disease, pulmonary sarcoidosis, and pneumoconiosis (eg, silicosis and asbestosis) [15, 28]. Further information on the ICD-10 codes used in defining CRD is available in Supplementary Table 2. Participants were enrolled during the 2023/2024 and 2024/2025 influenza seasons, defined in the study as between September and March [29]. According to the Chinese guidelines for influenza, both NAATs and rapid antigen tests (RATs) can be used for influenza diagnosis [30]. Therefore, the study outcomes were laboratory-confirmed influenza A/B virus by NAATs or RATs.

We screened the electronic health records of patients with chronic respiratory diseases (CRDs) who presented with an episode of acute respiratory infection (ARI), as defined by ICD-10 codes (see Supplementary Table 3). Influenza testing was not systematic but was performed at the discretion of the treating clinicians using RATs or NAATs. From this population, we included all patients with CRD who underwent such testing within 14 days of the ARI onset. The following patients were excluded: (1) those lacking an identification number to link vaccination information; (2) those whose influenza test was neither by RAT nor NAAT; and (3) those with duplicated influenza tests conducted within 14 days or outside the study period. If a patient had a new ARI episode at least 14 days after a previous one, they could be included again for that new episode. The study included subjects from outpatient, inpatient, and emergency departments.

Case and Control Definition

In the TND case-control study, laboratory-confirmed cases were defined as patients with CRD with medically attended ARI who tested positive for influenza by RAT or NAAT. In contrast, controls were patients with CRD with medically attended ARI who tested negative for influenza.

Exposure

Vaccination status was defined as documented receipt of at least 1 dose of influenza vaccine within the past 12 months and ≥14 days before sampling in the Shanghai immunization registry (covers all vaccination records in Shanghai). The 14-day interval allows for adaptive immune response maturation, and the 12-month limit was set because influenza vaccination is available year-round in Shanghai and serum antibody levels decline significantly 1 year post-vaccination [31, 32]. Subjects meeting these criteria were considered vaccinated; those without documentation or vaccinated <14 days before sampling were categorized as unvaccinated. Our analysis included all influenza vaccine types recommended by the Chinese guideline for the Northern Hemisphere: trivalent and quadrivalent inactivated vaccines, as well as the quadrivalent live-attenuated vaccine [33]. The specific intervals and doses of influenza vaccines for different age groups are recommended by the guideline presented in Supplementary Figure 1 [33].

Other Covariate Factors

Other key factors, such as age, sex, hospitalization status, influenza vaccination in the previous season (at least 1 dose of influenza vaccine within the past 13–24 months), history of pneumococcal vaccination, laboratory-confirmed previous influenza infection in the season, and specific CRD types, were all collected through the Health Big Data Platform and immunization registry records in Shanghai.

Sample Size

The sample size calculation for an unmatched case-control design using the Fleiss method can be used for TND [34]. Assuming 10% vaccination coverage among patients with CRD in controls (influenza-negative) [10] and an odds ratio (OR) of 0.54 reflecting influenza VE [12], we estimated 5.66% coverage in cases (influenza-positive). With 80% power to detect this effect at a 1-sided significant level of 0.05 and a 1:3 case-control ratio, a minimum required sample size was estimated at 471 cases and 1413 controls. A total sample size of 1884 subjects was required.

Statistical Analysis

Baseline characteristics were presented as median and interquartile range (IQR) for continuous variables or proportions for categorical variables. Group characteristics were compared using the chi-squared or Fisher's exact tests. Univariate and multivariate unconditional logistic regression models were used to estimate crude and adjusted ORs with 95% CIs, adjusting for age, sex, hospitalization status, influenza season, methods for laboratory-confirmed influenza, chronic respiratory disease types, and the month when specimens were collected in the multivariate models. VE was estimated by comparing vaccination coverage in influenza-positive cases and influenza-negative controls and calculated as 100 × (1-OR) [35, 36].

In addition to estimating the overall VE, subgroup analyses were conducted to assess the potential heterogeneity in VE across different strata of key variables. These variables included influenza seasons, influenza virus subtypes, age groups, hospitalization status, methods for laboratory-confirmed influenza, influenza vaccination in the previous season, pneumococcal vaccination, and the most common types of CRD.

We did sensitivity analyses to test the robustness of the results. Firstly, we used conditional logistic regression to validate our statistical assumptions on the estimates of ORs for influenza vaccination, matching each influenza-positive case with 3 negative controls by sex, age (±5 years), influenza season, and specimen collection month. Secondly, we evaluated the impacts of different criteria for classifying influenza season on the estimates of VEs, by defining influenza epidemics using 5% and 10% thresholds of influenza positive rate over 2 or more consecutive weeks [37] (Figure 1). Thirdly, we included subjects from both epidemic and non-epidemic periods (from 2023, week 23 to 2025, week 13) to assess the influences of potential misclassification of ILI during non-epidemic periods. Fourthly, to account for the waning immunity of vaccination [38], we performed analyses of VEs in subjects who were vaccinated <180 days before infection. Fifthly, since unvaccinated status relied solely on the linkage of documented influenza vaccination records in the Shanghai immunization registry, external vaccinations outside Shanghai might be missed in the system. To account for the potential bias of misclassifying vaccination status, we re-estimated VEs in subjects excluding those without vaccination records for both influenza and COVID-19. Sixthly, subjects sampled >3 days after their medical visit were excluded due to potential test result inaccuracy. Lastly, we only included subjects diagnosed by NAATs within 7 days to reduce potential bias caused by false-negative results.

All statistical analyses were conducted in R, version 4.1.0 (R Foundation). The $P < .05$ was considered statistically significant.

Ethical Considerations

The study was approved by the Medical Ethics Committee of the Chinese Academy of Medical Sciences and Peking Union Medical College (CAMS&PUMC-IEC-2024–068).

Reporting

The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) checklist for case-control studies was used to guide the transparent reporting of this TND case-control study (Supplementary Table 4).

RESULTS

Population Characteristics of Influenza Infection and Influenza Vaccination

During the 2023/2024 and 2024/2025 seasons, 10 711 subjects were included in the primary analysis (Supplementary

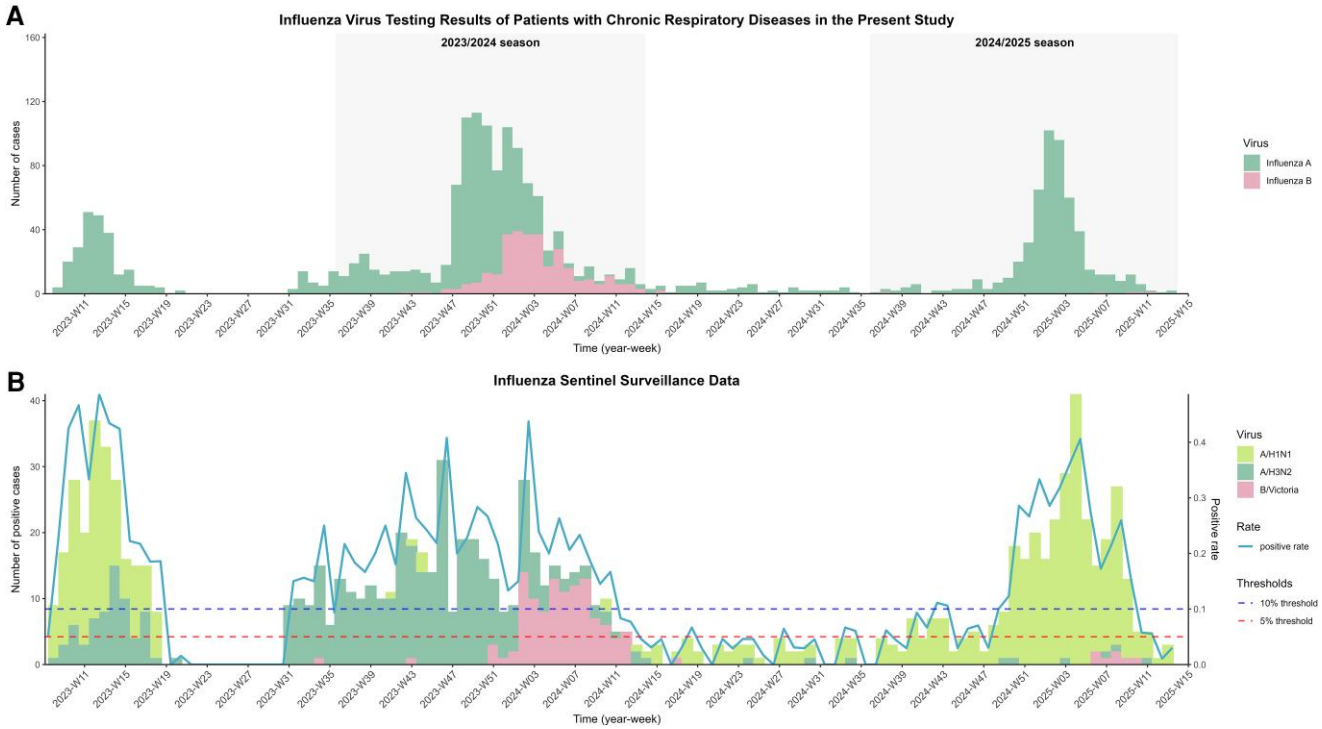

Figure 1. Comparison of the influenza cases between the present study and the sentinel surveillance system. *A*, Time series weekly number of influenza cases in patients with chronic respiratory disease, Pudong New Area, Shanghai, 2023–2025; *B*, Influenza sentinel surveillance data from the Chinese National Influenza Center (CNIC), Pudong New Area, Shanghai, 2023–2025. Gray shaded areas denote influenza seasons.

Figure 2). Of those, 1650 (15.4%) CRD with CRD tested positive for the influenza virus, and 9061 (84.6%) tested negative. Characteristics of these subjects are shown in Table 1. The overall median age was 53.3 years (IQR: 11.5–72.8 years). Most subjects (64.6%) were from the 2023/2024 season; 54.9% were males; and 46.3% aged 60 years or above. The majority (73.8%) were tested for influenza by RAT. Among the patients with CRD, 64.0% had chronic bronchitis, 20.2% had COPD, 8.9% had asthma, 6.3% had bronchiectasis, and 4.2% suffered from other CRDs (including interstitial-related respiratory diseases (3.3%), lung diseases caused by external agents [0.6%], and emphysema [0.3%]) (Table 1).

When compared with the control group, the patients with CRD in the case group were younger, with a higher proportion of males, and had a lower hospitalization rate. Moreover, the cases were less likely to be vaccinated with the influenza vaccine or the pneumococcal vaccine, and also less likely to have a laboratory-confirmed previous influenza infection during the season (Table 1).

In comparison with the unvaccinated group, the patients with CRD in the vaccinated group were younger and had a lower hospitalization rate. They were also more likely to have received the influenza vaccine in the previous season and the pneumococcal vaccine (Table 1).

Figure 1 presents the comparative time-series plots of influenza-positive case detections in our study and the corresponding CNIC influenza sentinel surveillance data. The cumulative influenza vaccine coverage alongside the daily incidence of laboratory-confirmed influenza cases are provided in Supplementary Figure 3.

Estimation of Influenza Vaccine Effectiveness

The combined VE for the study population was estimated at 44.23% (95% CI: 30.45–55.75). In the 2023/2024 season, it was 42.91% (95% CI: 27.13–55.78), and in the 2024/2025 season was 51.51% (95% CI: 17.70–73.61). The VE for influenza subtypes A and B were 38.49% (95% CI: 21.75–52.27) and 62.64% (95% CI: 39.91–78.28), respectively (Table 2).

Among hospitalized patients, the VE was 47.84% (95% CI: 1.15–75.74), while that was 45.32% (95% CI: 30.84–57.24) for nonhospitalized patients. Compared to the unvaccinated group, VE was lower among patients vaccinated in both the previous and current seasons (38.30%) than among those vaccinated only in the current season (49.81%). Patients who received pneumococcal vaccines had a higher VE (44.08%) than those without (36.71%). The VE varied by CRD types, being lowest in chronic bronchitis (42.93%) and highest in COPD (64.14%). The VE was higher for laboratory-confirmed influenza cases diagnosed using NAAT (57.20%) compared to RAT (43.61%) (Table 2).

Table 1. Demographic and Clinical Characteristics of Study Participants by Influenza Infection and Vaccination Status

Characteristics	Influenza-Positive Cases	Influenza-Negative Controls	*P*	Vaccinated[a]	Unvaccinated	*P*	Total
N	1650	9061		815	9896		10 711
Influenza season (%)			<.001			<.001	
2023/2024	1126 (68.2)	5788 (63.9)		632 (77.5)	6282 (63.5)		6914 (64.6)
2024/2025	524 (31.8)	3273 (36.1)		183 (22.5)	3614 (36.5)		3797 (35.4)
Sex (%)			.007			.246	
Male	956 (57.9)	4923 (54.3)		431 (52.9)	5448 (55.1)		5879 (54.9)
Female	694 (42.1)	4138 (45.7)		384 (47.1)	4448 (44.9)		4832 (45.1)
Age, years (median [IQR])	31.94 [9.68, 70.04]	56.36 [12.11, 73.15]	<.001	8.35 [5.01, 45.04]	57.11 [13.12, 73.19]	<.001	53.34 [11.47, 72.76]
Age group (%)			<.001			<.001	
<18 y	768 (46.5)	2753 (30.4)		580 (71.2)	2941 (29.7)		3521 (32.9)
18–59 y	259 (15.7)	1968 (21.7)		50 (6.1)	2177 (22.0)		2227 (20.8)
≥60 y	623 (37.8)	4340 (47.9)		185 (22.7)	4778 (48.3)		4963 (46.3)
Hospitalization (%)	467 (28.3)	2803 (30.9)	.035	110 (13.5)	3160 (31.9)	<.001	3270 (30.5)
Virus subtype (%)			<.001			.011	
Influenza A	1338 (81.1)	0 (0.0)		79 (9.7)	1259 (12.7)		1338 (12.5)
Influenza B	321 (18.9)	0 (0.0)		17 (2.1)	295 (3.0)		312 (2.9)
No influenza detected	0 (0.0)	9061 (100.0)		719 (88.2)	8342 (84.3)		9061 (84.6)
Method for lab-confirmed influenza (%)			.003			<.001	
NAAT	481 (29.2)	2324 (25.6)		109 (13.4)	2696 (27.2)		2805 (26.2)
RAT	1169 (70.8)	6737 (74.4)		706 (86.6)	7200 (72.8)		7906 (73.8)
Influenza vaccination in the previous season (%)	90 (5.5)	582 (6.4)	.151	405 (49.7)	267 (2.7)	<.001	672 (6.3)
Pneumococcal vaccination (%)	137 (8.3)	914 (10.1)	.028	290 (35.6)	761 (7.7)	<.001	1051 (9.8)
Lab-confirmed previous influenza in the season (%)	6 (0.4)	602 (6.6)	<.001	42 (5.2)	566 (5.7)	.553	608 (5.7)
Disease type (%)							
Chronic bronchitis	1155 (70.0)	5697 (62.9)	<.001	613 (75.2)	6239 (63.0)	<.001	6852 (64.0)
COPD	289 (17.5)	1867 (20.7)	.003	91 (11.2)	2074 (21.0)	<.001	2165 (20.2)
Asthma	112 (6.8)	841 (9.3)	.001	84 (10.3)	869 (8.8)	.160	953 (8.9)
Bronchiectasis	98 (5.9)	573 (6.3)	.591	24 (2.9)	647 (6.5)	<.001	671 (6.3)
Interstitial-related respiratory diseases	45 (2.7)	311 (3.4)	.163	12 (1.5)	344 (3.5)	.003	356 (3.3)
Lung diseases caused by external agents	6 (0.4)	63 (0.7)	.167	0 (0.0)	69 (0.7)	.030	69 (0.6)
Emphysema	4 (0.2)	26 (0.3)	.951	3 (0.4)	27 (0.3)	.881	30 (0.3)

Abbreviations: IQR, interquartile range; NAAT, nucleic acid amplification test; RAT, rapid antigen test; Lab, laboratory; COPD, chronic obstructive pulmonary disease.

[a]Subjects were considered vaccinated if they had documented records in the Shanghai immunization registry indicating receipt of at least 1 dose of influenza vaccine within the past 12 m and at least 14 d before sampling.

Sensitivity Analysis

The VE estimates from the sensitivity analyses were consistent with the combined VE in the main analysis. First, matching at a 1:3 ratio yielded similar results. Second, defining the influenza epidemic by the influenza-positive rate in CNIC data did not substantially change the VE. Third, including all subjects during the influenza epidemic and non-epidemic periods left the VE largely unchanged. Fourth, excluding subjects vaccinated >180 days before infection led to a slightly higher VE. Fifth, excluding subjects without influenza and COVID-19 vaccination records also yielded a marginally higher VE. Sixth, excluding subjects sampled >3 days after their medical visit resulted in a slightly higher VE than the main analysis. Lastly, only including subjects diagnosed by NAATs within 7 days led to a slightly higher VE (Table 3).

DISCUSSION

This is the first study to evaluate influenza VE among patients with CRD in China. Using a multicenter, TND case-control study in a real-world setting, we found that current influenza vaccines were effective in protecting patients with CRD from circulating strains, with a combined VE of 44.23% across 2 seasons (42.91% in 2023/2024% and 51.51% in 2024/2025).

Overall, our estimates of the VE for the influenza vaccine, both combined and by subtype, were consistent with previous studies conducted in the general population. For instance, in the 2023/2024 season, the overall VE was 49.4% in Ningbo (near Shanghai) [12] and 44.77% in Beijing (the largest city in Northern China) [13], which is comparable to our estimate of 42.91% for patients with CRD.

Table 2. Overall and Subgroup Analysis of the Crude and Adjusted Influenza Vaccine Effectiveness Against Infection

	Influenza-Positive Cases		Influenza-Negative Controls		Vaccine Effectiveness (%)	
Groups	No. Vaccinated[a]/Total	% Vaccinated	No. Vaccinated/Total	% Vaccinated	Crude (95% *CI*)	Adjusted[b] (95% *CI*)
Overall						
Season 1 and 2[c]	96/1650	5.8	719/9061	7.9	28.33 (11.17–42.79)	44.23 (30.45–55.75)
2023/2024 season	82/1126	7.3	550/5788	9.5	25.20 (5.33–41.61)	42.91 (27.13–55.78)
2024/2025 season	14/524	2.7	169/3273	5.2	49.58 (15.51–72.28)	51.51 (17.70–73.61)
Virus subtype						
Influenza A	79/1338	5.9	719/9061	7.9	27.20 (8.07–43.12)	38.49 (21.75–52.27)
Influenza B	17/312	5.4	719/9061	7.9	33.14 (−6.23–60.75)	62.64 (39.91–78.28)
Age group						
<18 y	80/768	10.4	500/2753	18.2	47.60 (33.03–59.49)	39.55 (22.09–53.61)
18–59 y	5/259	1.9	45/1968	2.3	15.58 (−94.70–71.06)	12.55 (−104.26–70.09)
≥60 y	11/623	1.8	174/4340	4.0	56.97 (24.12–78.09)	57.04 (23.87–78.21)
Hospitalization status						
Yes	9/467	1.9	101/2803	3.6	47.43 (1.01–75.45)	47.84 (1.15–75.74)
No	87/1183	7.4	618/6258	9.9	27.56 (9.00–43.00)	45.32 (30.84–57.24)
2 consecutive seasons' vaccination history						
Current and previous	51/1605	3.2	354/8696	4.1	22.66 (−3.22–43.25)	38.30 (16.84–54.74)
Current only	45/1599	2.8	365/8707	4.2	33.82 (10.40–52.29)	49.81 (31.57–64.04)
Pneumococcal vaccination						
Yes	26/137	19.0	264/914	28.9	42.33 (10.89–63.92)	44.08 (12.79–65.29)
No	70/1513	4.6	455/8147	5.6	17.99 (−5.39–37.12	36.71 (18.11–51.77)
Chronic respiratory disease type						
Chronic bronchitis	85/1155	7.4	528/5697	9.3	22.23 (1.84–39.10)	42.93 (27.47–55.60)
COPD	5/289	1.7	86/1876	4.6	63.36 (17.70–87.18)	64.14 (18.93–87.50)
Asthma	4/112	3.6	80/841	9.5	64.77 (13.18–88.40)	56.67 (−9.59–87.16)
Methods for laboratory-confirmed influenza						
NAAT	9/481	1.9	100/2324	4.3	57.59 (20.11–80.20)	57.20 (18.99–80.08)
RAT	87/1169	7.4	619/6737	9.2	20.53 (.18–37.46)	43.61 (28.65–55.93)

Vaccine effectiveness was estimated by comparing the vaccination coverage in influenza-positive cases and influenza-negative controls and calculated as 100 × (1—odds ratio) in logistic regression models.

Abbreviations: CI, confidence interval; NAAT, nucleic acid amplification test; RAT, rapid antigen test; COPD, chronic obstructive pulmonary disease.

[a]Subjects were considered vaccinated if they had documented records in the Shanghai immunization registry indicating receipt of at least 1 dose of influenza vaccine within the past 12 m and at least 14 days before sampling.

[b]Adjusted for age, sex, hospitalization status, influenza season, methods for laboratory-confirmed influenza, chronic respiratory diseases and the month when specimens were collected.

[c]Season 1 was 2023/2024; season 2 was 2024/2025.

Our study revealed a notable difference in VE between influenza A and B. The VE against influenza B (62.64%) was higher than that against influenza A (38.49%), which is in line with previous studies in the general population [12, 13] and the elderly [39] in China. This difference may be attributed to influenza vaccines providing substantial protection against A/H1N1 and type B influenza but exhibiting reduced protection against A/H3N2 [40]. The lower VE in 2023/2024 can be explained by the predominant strains (A/H3N2 and type B) having relatively less vaccine protection, while the more vaccine-effective A/H1N1 was the major strain in 2024/2025 (Figure 1*B*). Despite seasonal variations, our results show the influenza vaccine provides consistent, moderate protection for patients with CRD across different strains and seasons, highlighting its continued importance for this vulnerable group. Furthermore, the timing of the vaccination campaign in 2024/2025 may also be part of the reason why the VE was higher than in 2023/2024 (Supplementary Figure 3). It is essential to estimate the burden averted by vaccination and translate VE into public health impact. This approach moves beyond VE by integrating key parameters: vaccination coverage, timing, and speed. When combined with VE estimates (like those from our study) and baseline disease rates, these data can model the tangible benefits of a vaccination program, specifically, the number of lives saved, hospitalizations averted, and absenteeism prevented. Generating such estimates provides Ministries of Health and Finance with critical evidence on the value of vaccination.

We observed significant variation in influenza VE across age groups among patients with CRD. This may be due to the markedly low vaccine coverage (2%) in the 18–59-year CRD subgroup and limited sample size compared to the other age groups [41]. The differences across age groups also explain why our models exhibited greater robustness when age was adjusted for or matched within 5 years. The lack of statistically significant protection in the asthma patients may also be influenced by the low vaccination rate and limited sample sizes.

Table 3. Sensitivity Analysis of the Crude and Adjusted Influenza Vaccine Effectiveness Against Infection

	Influenza-Positive Cases		Influenza-Negative Controls		Vaccine Effectiveness (%)	
Number of Subjects Included	No. Vaccinated[a]/Total	% Vaccinated	No. Vaccinated/Total	% Vaccinated	Crude (95% *CI*)	Adjusted[b] (95% *CI*)
Sensitivity analysis 1: a 1:3 ratio non-replacement matching by sex, age (±5 y), influenza season, and the month when specimens were collected						
6171	96/1650	5.8	409/4521	9.0	45.70 (30.09–57.83)	43.43 (26.82–56.27)
Sensitivity analysis 2: Defining the start and end of an influenza epidemic by the influenza positive rate of ILI sentinel surveillance						
8789[c]	88/1384	6.4	654/7405	8.8	29.91 (12.24–44.67)	46.96 (33.07–58.43)
7784[d]	85/1340	6.3	565/6444	8.8	29.53 (11.27–44.69)	46.42 (32.03–58.23)
Sensitivity analysis 3: Including all subjects during the influenza epidemic and non-epidemic periods[e]						
13 307	106/1771	6.0	915/11 536	7.9	26.10 (9.48–40.27)	43.24 (30.06–54.37)
Sensitivity analysis 4: Excluding subjects vaccinated more than 180 d before infection						
10 562	75/1629	4.6	591/8933	6.6	31.88 (13.44–47.14)	46.33 (31.37–58.60)
Sensitivity analysis 5: Excluding subjects without influenza and COVID-19 vaccination records						
5901	96/901	10.7	719/5000	14.4	28.99 (11.43–43.65)	52.53 (39.89–62.86)
Sensitivity analysis 6: Excluding subjects whose sampling occurred more than 3 d after their medical visit						
10 682	95/1648	5.8	718/9034	7.9	29.15 (12.11–43.51)	44.95 (31.28–56.36)
Sensitivity analysis 7: Including subjects whose sampling occurred within 7 d after their medical visit and diagnosed by NAAT						
2803	9/481	1.9	100/2322	4.3	57.63 (20.19–80.22)	57.24 (19.08–80.10)

Vaccine effectiveness was estimated by comparing the vaccination coverage in influenza-positive cases and influenza-negative controls and calculated as 100 × (1—odds ratio) in logistic regression models.

Abbreviations: CI, confidence interval; ILI, influenza-like illness; NAAT, nucleic acid amplification test.

[a]Subjects were considered vaccinated if they had documented records in the Shanghai immunization registry indicating receipt of at least 1 dose of influenza vaccine within the past 12 m and at least 14 d before sampling.

[b]Adjusted for age, sex, hospitalization status, influenza season, methods for laboratory-confirmed influenza, chronic respiratory diseases, and the month when specimens were collected.

[c]5% threshold of the influenza positive rate for 2 or more consecutive weeks. The start time was week 31, 2023, and the end time was week 12, 2024 for the 2023/2024 season; the start time was week 37, 2024, and the end time was week 13, 2025 for the 2024/2025 season.

[d]10% threshold of the influenza positive rate for 2 or more consecutive weeks. The start time was week 31, 2023, and the end time was week 10, 2024 for the 2023/2024 season; the start time was week 42, 2024, and the end time was week 9, 2025 for the 2024/2025 season.

[e]All subjects from 2023, week 23 to 2025, week 13 were included in this sensitivity analysis.

Notably, the VE among patients with CRD vaccinated in both previous and current seasons was lower than those vaccinated only in the current season (38.30% vs 49.81%). The result was consistent with the previous observational study of the general population [42] and a meta-analysis [43]. The meta-analysis indicated that repeated vaccination may reduce VE against influenza A/H3N2, whereas the reduction in VE against influenza B was comparatively small. A study showed that antibody responses to A/H3N2 can be blunted with each additional vaccination received [44]. This may be attributed to both high viral heterogeneity complicating antigen selection for influenza A/H3N2 [45] and antigenic alterations in egg-based production [46]. While cell-grown or recombinant vaccines show promise, insufficient data currently exist to determine whether they mitigate the attenuating effects of repeated vaccination [43].

In Shanghai, people aged 60 or above can get a 23-valent pneumococcal vaccination for free. However, all people must pay out of pocket for the influenza vaccination. Our findings indicated that subjects who received both pneumococcal and influenza vaccines had a higher influenza VE than those vaccinated with the influenza vaccine only. Prior evidence demonstrates the additive effect of the pneumococcal vaccine and the influenza vaccine on acute exacerbation in patients with CRD [47].

The strengths of our study lie in the following aspects. First, this study evaluated the influenza VE among patients with CRD using multicenter data with a large sample size. After COVID-19, the seasonal circulation patterns of influenza and people's health awareness changed substantially. Our study can provide the latest scientific data for the public. Second, the study employed a TND case-control approach, which is a feasible and the most efficient method currently available for the influenza VE assessment in the real-world setting. When compared to traditional cohort and case-control designs, the TND design yields the most accurate estimates by substantially reducing confounding from healthcare-seeking behavior [25, 48, 49]. Third, we defined the cases, controls, baseline characteristics, and vaccination exposure status by medical records or immunization registry records, which reduced recall bias and misclassification bias. Finally, we evaluated the VE among patients with CRD in 2 consecutive epidemic seasons across different influenza circulating strains, which provides critical insights into the stability of immune protection among patients with CRD.

This study has some limitations. Firstly, low coverage (41%) of inpatient data in the Big Data Platform is a limitation that may miss some severe cases. However, the TND case-control study inherently mitigates this bias as it equally affects both cases and controls selected from the same patient population.

Secondly, although the guideline-recommended RAT was used to confirm most influenza cases due to its convenience, its low specificity may underestimate the overall influenza VE. Nevertheless, we have accounted for this potential bias in our model, and the sensitivity analysis by only using NAATs does not substantially change our results and conclusions. Thirdly, low vaccine coverage and inadequate sample sizes within some subgroups compromised the precision of influenza VE estimates. Future studies should boost vaccination rates in different populations and collaborate across stakeholders for more accurate estimates. Finally, the definition of the unvaccinated status of influenza relied solely on the Shanghai immunization registry, which may have caused misclassification and underestimated VE, as 35% of subjects had no records of any vaccination. However, we performed a sensitivity analysis excluding individuals lacking both influenza and COVID-19 vaccination records, which mitigated this potential bias.

In summary, current influenza vaccines effectively protect patients with CRD against circulating strains. Our findings offer robust real-world evidence that influenza vaccination provides consistent and moderate protection for patients with CRD against medically attended influenza, regardless of the dominant circulating subtypes. Nevertheless, vaccination coverage remains suboptimal, underscoring the need to improve annual influenza vaccination uptake among patients with CRD.

Supplementary Data

Supplementary materials are available at *Open Forum Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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Authors' contributions. Y. L., data curation, formal analysis, methodology, visualization and original draft writing; J. Y., methodology, validation and revised the draft; C. Y. and Z. L., conceptualization, methodology, and supervision; X. Y. and L. X., data curation and formal analysis; E. G., H. X. and Q. H., validation, revised the draft and supervision; W. Y., validation, supervision and funding acquisition; L. H., supervision and project administration. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

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Potential conflicts of interest. All authors: No reported conflicts.

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